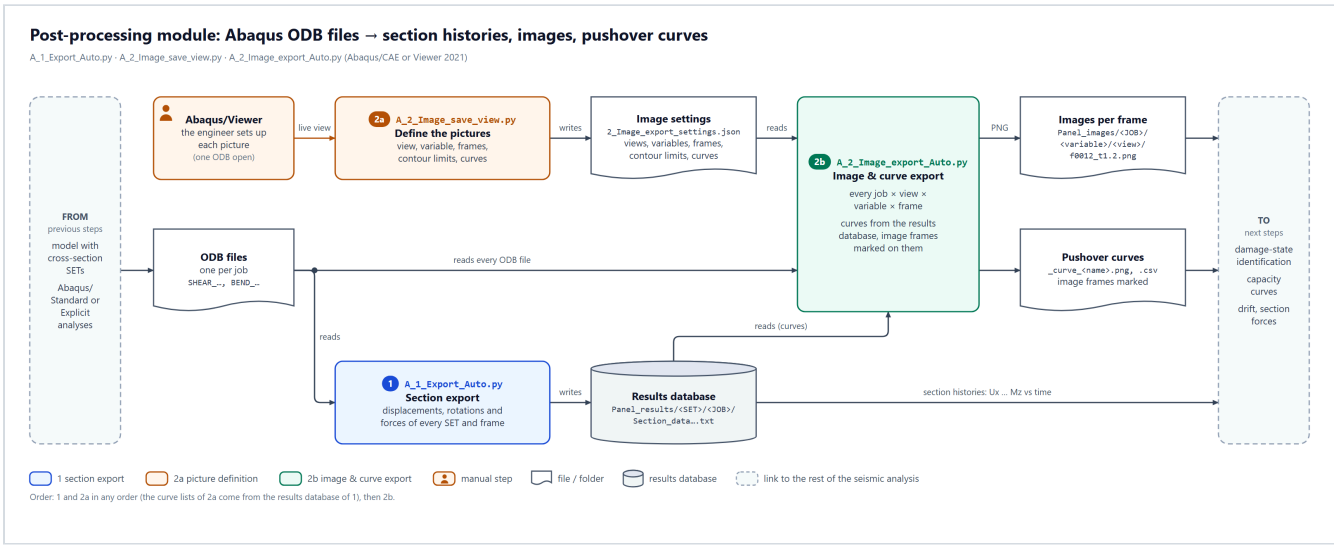
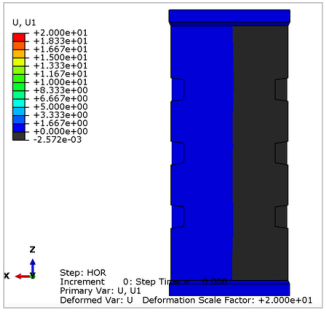
TUTORIAL

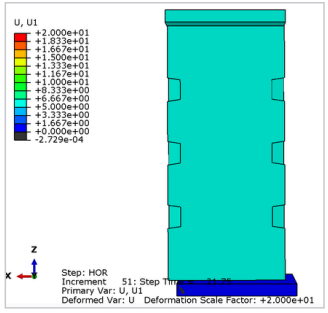
Abaqus scripts for output processing

From the model to section forces, pictures and pushover curves: preparing the model, running the analyses, exporting cross-sections, images and curves in Abaqus/CAE and Abaqus/Viewer.

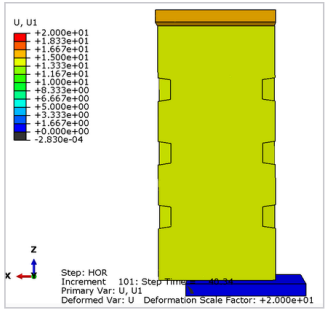




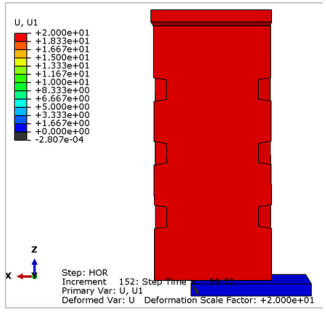
f0000_t0.png
frame 0



f0051_t21.7527.png
frame 51



f0101_t40.3361.png
frame 101



f0152_t59.7221.png
frame 152

Repository	github.com/RomanJasen/Abaqus-scripts-for-output-processing
Scripts	Abaqus/CAE and Abaqus/Viewer 2021 (Python 2.7), Windows 10
Example	masonry wall benchmark with 16 cross-sections (implicit and explicit pushover analyses)
Version	November 2026

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0 Overview and conventions

The scripts automate a numerical (seismic, pushover) analysis of structures modelled with **volumetric (solid) or shell elements** in Abaqus: the model is prepared once, many analyses are run automatically, and the forces and displacements of chosen cross-sections, pictures of the results and pushover curves are exported for every analysis. They were developed for masonry walls and buildings, but work wherever section forces and displacements are needed.

Step	Folder → script(s)	What it does
1	`1_model_preparation` `prepare_model_for_export.py`	checks the cross-section SETs of the model (chapter 1)
2	`2_run_analyses` `run_pushover_jobs.py`, `run_parameter_study.py`	runs the pushover analyses of the chosen models, or a parameter study (chapter 2)
3	`3_export_and_image_export` `A_1_Export_Auto.py`	displacements, rotations and forces of every cross-section and frame → results database (chapter 3)
4	`3_export_and_image_export` `A_2_Image_save_view.py`, `A_2_Image_export_Auto.py`	pictures defined once in Abaqus/Viewer and printed for every analysis and frame; pushover curves with the frames of the pictures marked (chapters 4 and 5)

Conventions shared by all scripts

- **Run them from Abaqus:** *File* > *Run Script...* in Abaqus/CAE (steps 1 and 2, with the model database open) or in Abaqus/CAE or Viewer (steps 3 and 4). No other software is needed: the plots use the matplotlib that comes with Abaqus.
- **INPUT block:** every script starts with a block of settings marked `INPUT`, each with a comment. Most scripts also show a dialog with the same settings before they start.
- **Settings are remembered** by the export scripts (steps 3 and 4) in `1_Export_Auto_settings.json` and `2_Image_export_settings.json` next to the scripts. A change in the dialog, in the JSON file or in the INPUT block is used; the latest change wins. Delete the JSON file to go back to the INPUT block.
- **Progress is logged** to a `*_log.txt` file. Abaqus often shows the messages of a script only when it has finished; open the log file to see how far a long run is.
- **Nothing is done twice:** analyses whose ODB exists, sections already in the results database and existing images are skipped, so an interrupted run is continued by running the script again.
- **The A_ files belong together:** to use steps 3 and 4 in a folder with ODB files, copy every file and folder whose name starts with `A_` into it (chapter 3.1).

The example of this tutorial

All figures come from the benchmark model of the repository's videos: a masonry wall (`PANEL_S-1`, 1200 × 2600 × 150 mm, C3D8R elements of about 50 mm) between a bottom beam and a top loading beam, with 16 cross-section SETs. The pictures, the results database and the curves of chapters 3–5 were produced by the scripts of this repository on the implicit analysis `SHEAR_key_S_FR05_FORCE_34272` (vertical force 34.3 kN, pushover to 20 mm); one curve comes from the explicit analysis `SHEAR_KEY_S_FR05_EX_172699`.

> **Check your results.** As with any automated post-processing, verify the exported forces and displacements of a few sections before relying on them: chapter 1.4 shows a mistake that gives plausible-looking but wrong forces, and chapter 3.4 how to check a section.

1 Preparing the model

1.1 What a cross-section SET is

A cross-section is a cut through the model at which the forces and displacements are wanted, for example the bottom and the top of a pier. For the scripts it is an **assembly-level SET** that contains

- a **node set: the nodes on the cut**, used for the displacements and rotations of the section and as the points at which the forces are summed, and
- an **element set: the elements on ONE side of the cut** (one layer touching the cut is enough), the "free body" whose forces at the cut nodes give the section forces,

both under the same name. Figure 1 shows such a SET of the benchmark wall.

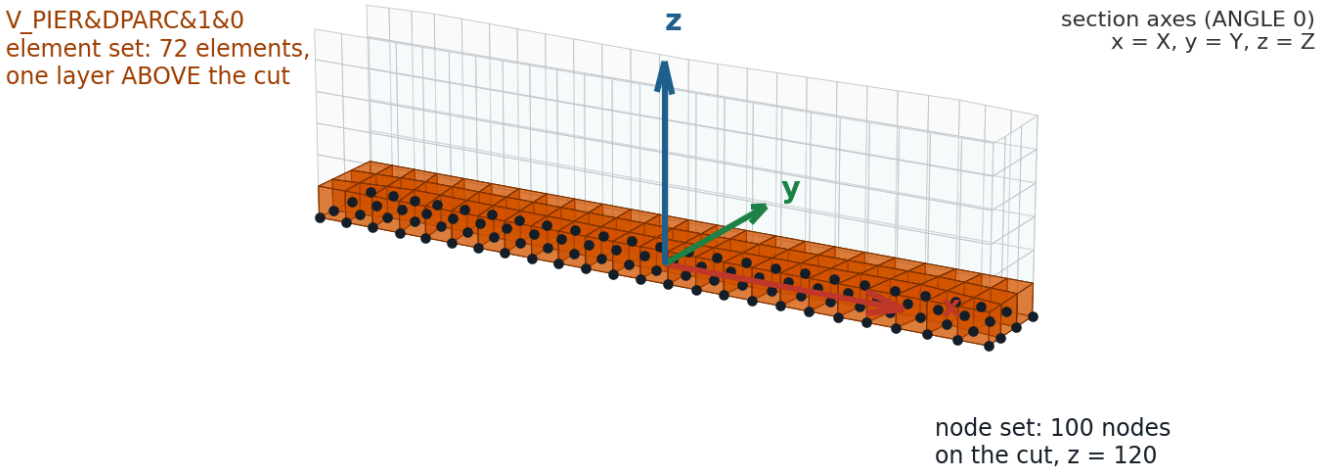


Figure 1. SET `V_PIER&DPARC&1&0` at the bottom of the wall: the node set holds the 100 nodes of the cut (z = 120 mm, black dots), the element set the 72 elements of the layer above it (orange). The bottom beam below the cut is not part of the SET. Arrows: the section axes for ANGLE 0.

1.2 Naming: TYPE&LABEL&INDEX&ANGLE

The name of a SET tells the scripts how to treat it. It has four parts separated by `&`:

V_PIER	&	DPARC	&	1	&	0
TYPE: kind of section		LABEL: free text, e.g. the position		INDEX: number, e.g. 1 = bottom, 2 = top		ANGLE: turn of the axes about global Z [°]

TYPE	geometry	section axes	forces	use
`V_PIER` , `V_SPEC`	solid elements	pier	yes	horizontal cuts of walls and piers
`S_PIER`	shell (line of nodes)	pier	yes	the same in shell models
`V_SPANDREL` , `S_SPANDREL`	solid / shell	spandrel	yes	vertical cuts of spandrels and beams
`S_DRIFT`	point (node average)	global	no	storey displacements (drift)
`SRIFT`	solid elements	pier	no	displacements and rotations only

- INDEX and ANGLE must be numbers; the SET name must not contain other `&` characters.

- Only SETs whose name contains one of `PIER`, `SPEC`, `SPANDREL`, `DRIFT` or `SRIFT` are exported; all other sets of the model are ignored. Helper sets must therefore not contain these words.
- Abaqus writes set names in capitals into the ODB; the scripts compare names without regard to case.

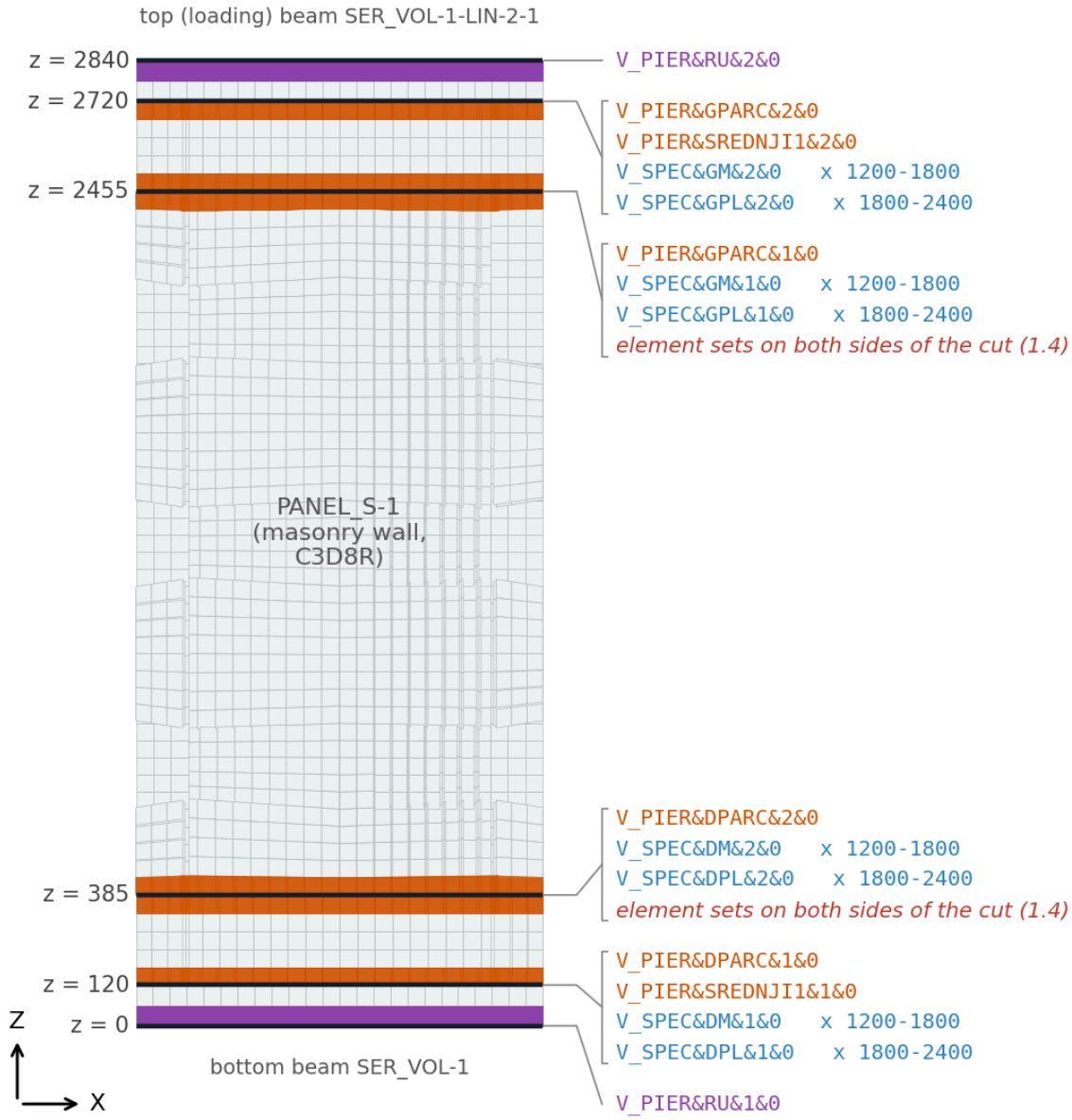


Figure 2. The 16 SETs of the benchmark wall in a front view. Several SETs share a cut: the `V_PIER` SETs span the whole wall, the `V_SPEC` SETs its left or right half; `RU` are the bottom of the bottom beam and the top of the loading beam. The two cuts in red italics are discussed in 1.4.

1.3 Section axes, components and signs

Displacements and forces are written in the **section axes** x, y, z, which depend on the TYPE and ANGLE (Figure 3). For a pier, z is the vertical axis (normal to the horizontal cut) and ANGLE turns x and y about it; ANGLE 0 gives x = global X, y = global Y. For a spandrel the cut is vertical: z is the beam axis, x points down.

PIER, ANGLE 0
e.g. V_PIER&RU&2&0

PIER, ANGLE 90
e.g. V_PIER&W1&1&90

SPANDREL, ANGLE 0
e.g. V_SPANDREL&S1&1&0

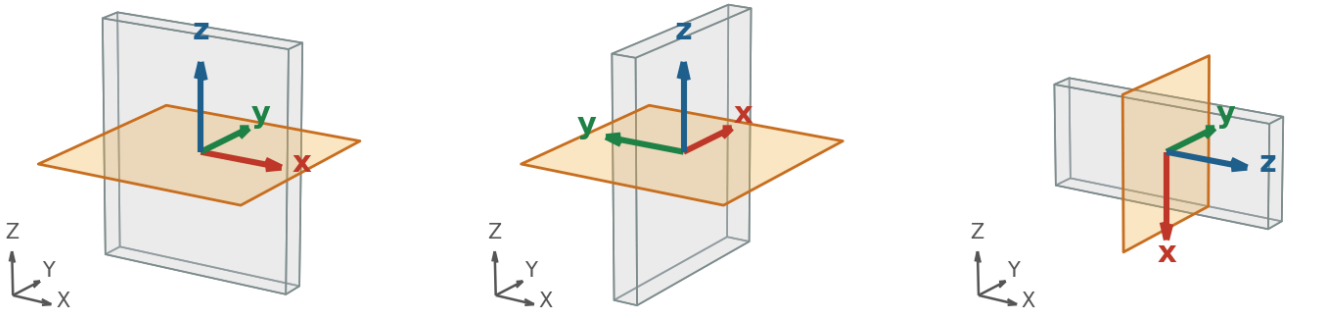


Figure 3. Section axes (x red, y green, z blue) and the cut (orange) of a pier with ANGLE 0 and 90 and of a spandrel. The small triads are the global axes X, Y, Z.

columns	meaning (pier with ANGLE 0, wall in the X–Z plane)
`Ux Uy Uz`	displacement of the section centre along x, y, z: Ux in-plane, Uy out of plane, Uz vertical
`fix fiy fiz`	rotation about x, y, z [rad]: fiy is the in-plane rotation (bending), fiz the twist
`Fx Fy Fz`	forces along x, y, z: Fx in-plane shear, Fy out-of-plane shear, Fz normal (axial) force
`Mx My Mz`	moments about x, y, z: My in-plane bending moment, Mx out-of-plane bending, Mz torsion

Signs of the forces. The forces are those that the rest of the model exerts *on the element set*. The same internal force therefore has the opposite sign when the element set lies on the other side of the cut. In the example (Figure 9), the top section `V_PIER&RU&2&0` (elements below the cut) gives Fz = −34.8 kN and a peak Fx = +23.8 kN, the base section `V_PIER&RU&1&0` (elements above the cut) Fz = +48.4 kN and Fx = −23.7 kN: both are compression and the same shear (the base also carries the self-weight). Put the element sets on the same side for all sections of a kind, or correct the sign in the post-processing (e.g. `"factor": -1` of a curve, chapter 4).

Results exported before 28 September 2026 were written with axes turned by 180° for odd INDEX; for those SETs Fx, Fy, Mx and My have the opposite sign.

1.4 Correct and wrong cuts

The element set must contain elements of **one side only**. If it holds the elements of both sides, the forces of the two sides cancel and the section reports nearly zero. The displacements of such a SET are still correct, because they only use the nodes. Both scripts check this: `prepare_model_for_export.py` in the model before the analyses (1.6), and the export for every SET it exports – a WARNING line in its log, a summary at the end of the log and a message box. For the benchmark:

```
WARNING: 6 SET(s) have elements on BOTH sides of the cut - their exported forces
         are wrong (about zero), the displacements are correct:
  V_PIER&DPARC&2&0, V_PIER&GPARC&1&0, V_SPEC&DM&2&0, V_SPEC&DPL&2&0, V_SPEC&GM&1&0, V_SPEC&GPL&1&0
  Fix the element sets (elements of one side only, see 1_model_preparation/README.md).
```

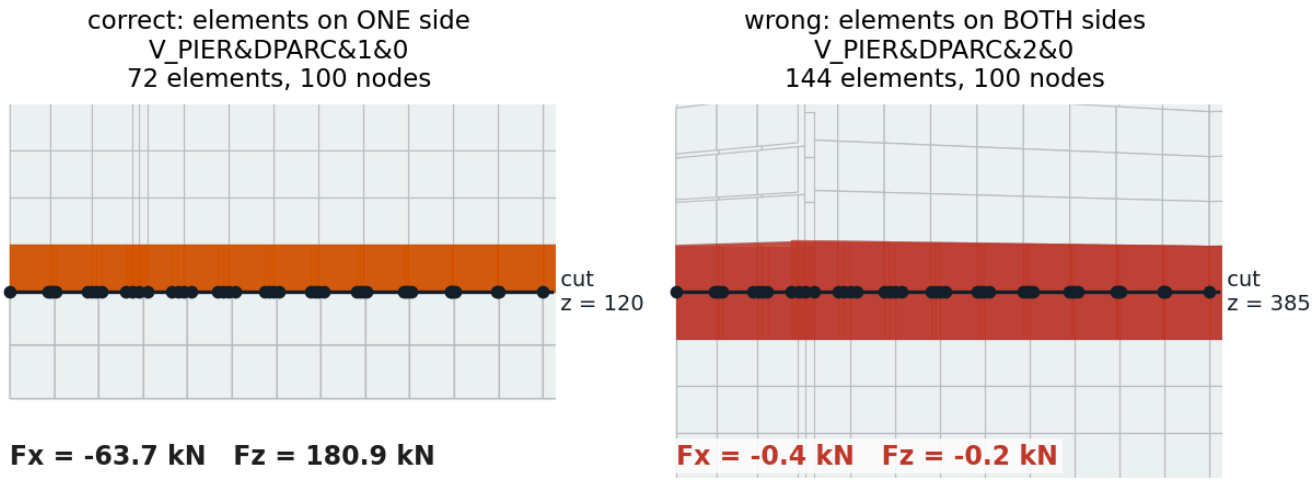


Figure 4. Two SETs of the benchmark (left half of the wall, front view) and their exported forces at step time 2.0 of the explicit job `SHEAR_KEY_S_FR05_EX_172699`. Left: the cut at z = 120 lies on the boundary of the wall; the SET has the elements of one side. Right: the cut at z = 385 lies on a partition face inside the wall; the SET has the elements of both sides and gives Fx = −0.4 kN instead of about 64 kN.

In the benchmark this concerns the SETs at z = 385 and z = 2455 (Figure 2): they were made as geometry sets of partition faces inside the wall, and Abaqus puts the elements of both sides of such a face into the set. Checklist for every SET:

- **node set:** all nodes of the cut and only those (one plane of nodes; for shells one line);
- **element set:** elements touching the cut, all on the same side (more layers on that side do no harm);
- **at a contact or tie interface** (two parts with their own nodes) take the nodes and the elements of the same part;
- **quick check:** the normal force Fz of a horizontal cut through the whole wall must equal the vertical load above it (plus self-weight). A value near zero means elements on both sides.

1.5 Creating the SETs in Abaqus/CAE

Abaqus/CAE writes a SET with nodes and elements under one name in two ways. Both were checked with Abaqus/CAE 2021, and both give an `*Nset` and an `*Elset` of that name in the input file:

a) Cut on the boundary of a part: geometry set of the face

1. Partition the part so that the cut is a face, and mesh it.
2. In the Assembly module: *Tools* > *Set* > *Create...*, name e.g. `V_PIER&DPARC&1&0`, type *Geometry*, select the face of the cut, *Done*.

Abaqus stores the nodes of the face and the elements next to it. This is correct only if all those elements are on one side, i.e. the face is on the boundary of the part instance (like the faces between the wall and the beams in the benchmark). For a face inside a part use b).

b) Cut inside a part: node set + element set combined

1. *Tools* > *Set* > *Create...*, name `cut_nodes`, type *Node*: select the nodes of the cut (easiest in a view looking along the cut, with a box selection, and with a display group showing only the region: *Tools* > *Display Group*).
2. *Tools* > *Set* > *Create...*, name `cut_elements`, type *Element*: select the layer of elements on one side of the cut.
3. Combine both into the section SET with one command in the kernel command line of Abaqus/CAE (click the `>>>` button left of the message area below the viewport):

```
a = mdb.models['Model-1'].rootAssembly
a.SetByBoolean(name='V_PIER&DPARC&2&0', operation=UNION,
               sets=(a.sets['cut_nodes'], a.sets['cut_elements']))
```

Repeat for every section (replacing `cut_nodes` and `cut_elements`). The helper sets do not contain a section TYPE and are ignored by the export; delete them at the end if you like.

> **Tip.** Check a SET in Abaqus/Viewer before the analyses are run: *Tools* > *Display Group* > *Create*, item *Elements*, method *Element sets*, select the SET and *Replace*. You must see one layer of elements on one side of the cut.

1.6 Field output and the check script

field output variable	needed for
`U`	displacements and rotations of the sections
`NFORC`	section forces (both the free-body cut and the NFORC method)
`UR`	rotation of shell sections about their own line (shell models only)
e.g. `DAMAGEC`, `DAMAGET`, `PEEQ`, `S`	the variables of the pictures

More output frames give more points on the curves and more frames to choose for the pictures.

`1_model_preparation/prepare_model_for_export.py` (Abaqus/CAE, model database open) lists the section SETs of one or all models, checks their names and whether the element set of every SET with forces lies on one side of the cut (the model must be meshed); optionally it sets the variables of the field output request `F-Output-1` (`FIELD_OUTPUT` in its INPUT block) and creates surfaces `SURF_<SET>` for the earlier scripts and videos (not needed by the current export). `convert_models_to_explicit_EXAMPLE.py` is a recorded example of converting the benchmark models to Abaqus/Explicit; record the same actions for your own model and use it as a template.

2 Running the analyses

Both scripts of `2_run_analyses` run in Abaqus/CAE with the model database open (*File* > *Run Script*...). The ODB files are written to the work directory (*File* > *Set Work Directory*...).

workflow	script	runs
A – pushover analyses	`run_pushover_jobs.py`	the chosen models at every vertical force
B – parameter study	`run_parameter_study.py`	the models with every value of a contact or CDP-material parameter

2.1 Workflow A: pushover analyses (run_pushover_jobs.py)

For every chosen model and every vertical force the script sets the force, creates the job `<model>_EX_<force>` and submits it, a given number of analyses at a time, each on a given number of cores. A dialog shows the INPUT block and lets you change it:

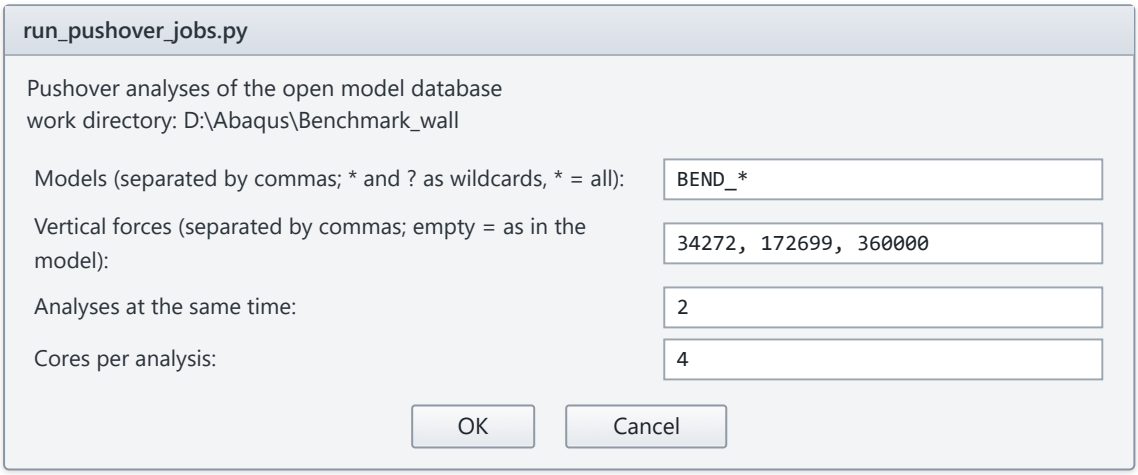


Figure 5. First dialog (Abaqus' own input dialog; the texts are those of the script, the values typed by the user).

field	meaning
Models	model names of the open .cae, separated by commas; `*` and `?` as wildcards (not case sensitive): `*` = all models, `BEND_*` = all models starting with BEND_
Vertical forces	one analysis per force, in the force unit of the model; the concentrated force `VERT_FORCE` gets `cf3 = -force` (downwards) in the steps `VERT` and `HOR`. Empty: the models are run with the force they have (job name `<model>_EX`).
Analyses at the same time	the script submits this many analyses; before the next one it waits for the oldest running analysis (analyses finished meanwhile also free their place)
Cores per analysis	`numCpus` and `numDomains` of every job. Analyses at a time × cores should not exceed the cores (and licence tokens) of the computer.

A second dialog lists what will be done and asks before anything in the model is changed:

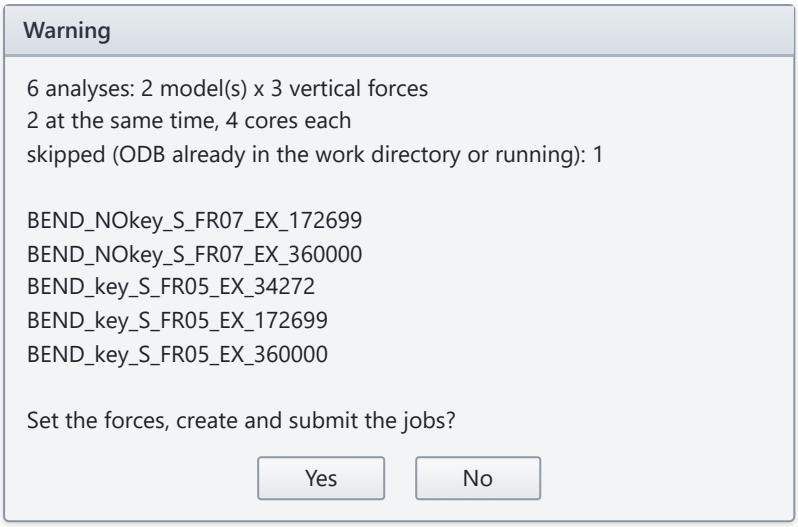


Figure 6. Confirmation: `BEND_NOkey_S_FR07_EX_34272.odb` already exists and is skipped.

Progress goes to `run_pushover_jobs_log.txt` in the work directory:

```
2026-11-12 10:15:43  skipped (BEND_NOkey_S_FR07_EX_34272.odb exists): BEND_NOkey_S_FR07_EX_34272
2026-11-12 10:15:44  submitted: BEND_NOkey_S_FR07_EX_172699 (1 running)
2026-11-12 10:15:46  submitted: BEND_NOkey_S_FR07_EX_360000 (2 running)
2026-11-12 10:15:46  waiting for BEND_NOkey_S_FR07_EX_172699
2026-11-12 11:02:17  finished: BEND_NOkey_S_FR07_EX_172699 (COMPLETED)
2026-11-12 11:02:19  submitted: BEND_key_S_FR05_EX_34272 (2 running)
...
2026-11-12 12:31:05  DONE: 5 submitted, 1 skipped, 0 failed - still running: BEND_key_S_FR05_EX_172699,
                     BEND_key_S_FR05_EX_360000 (see the Job Manager)
```

- Abaqus/CAE stays busy until the last analysis has been *submitted*; the last ones then run on (`WAIT_AT_END = True` also waits for them).
- Jobs whose `<job>.odb` or `<job>.lck` exists are skipped: run the script again to continue an interrupted series; delete the ODB of a failed analysis to repeat it.
- INPUT block only: `LOAD_NAME`, `LOAD_STEPS`, the job name pattern `JOB_NAME`, `JOB_SETTINGS` (memory, explicit precision, ...), `SUBMIT = False` (only write the input files `<job>.inp`, e.g. for another computer) and `ASK = False` (no dialogs, e.g. `abaqus cae noGUI=run_pushover_jobs.py`).

2.2 Workflow B: parameter study (run_parameter_study.py)

For every model and every value of the studied parameter the model is changed, a job is created and submitted. The study, the models and the values are set in the INPUT block:

```
STUDY = "material"                  # INPUT: "contact" | "material"
MODELS = ["BEND_NOkey_S_FR07"]      # INPUT: models of the open .cae to run
JOB_PREFIXES = None                 # INPUT: None = the model names; or one prefix per model
# -- contact
CONTACT_PROPERTY = "Simple_contact"   # interaction property of the model
CONTACT_FRICTION = 0.5
CONTACT_DAMPING = [0, 0.0001, 0.001, 0.01]
CONTACT_PRESSURE = [1, 5, 10, 100, 1000]   # exponential pressure-overclosure
# -- material
MATERIAL_NAME = "BETON"             # material of the model that gets the CDP data
# -- jobs
SUBMIT = True;  JOBS_AT_ONCE = 2;  CORES_PER_JOB = 4
```

STUDY	changed in the model	job name
`contact`	interaction property: friction, exponential pressure-overclosure, damping (every damping × every pressure)	`<prefix>_Damping_<d>pressure<p>`
`material`	material `BETON` : elasticity and the CDP tables, one job per material of `material_CDP`	`<model>_MAT_<material>`

Skipping of finished jobs, analyses at a time, cores and `SUBMIT = False` work as in workflow A; the log file is `run_parameter_study_log.txt` . A new kind of study is added in chapter 6.1.

2.3 CDP materials

The material study reads concrete damaged plasticity (CDP) materials from the folder `2_run_analyses/material_CDP` , given in one of two ways:

input folder	format
`A_Material_input_indirect`	one line per material, tab separated: `name E v s_max e_smax e_u (not used) Sigma_T GFI` . The compression curve follows the Eurocode formula $\sigma = s_{max} (k\eta - \eta^2) / (1 + (k - 2)\eta)$, the tension is given by the tensile strength and the fracture energy (Abaqus type GFI).
`A_material_input_direct`	one text file per material with the tables themselves, e.g. calibrated in a spreadsheet (layout: `Example_direct.txt` ; the layout is checked).

```
# A_Material_input_indirect/Example_EC_materials.txt (made-up example values)
        Example_EC_C20  30000   0.2     20      0.002   0.0035  0.06    2.0     0.1
        Example_EC_C30  33000   0.2     30      0.0022  0.0035  0.07    2.9     0.13
```

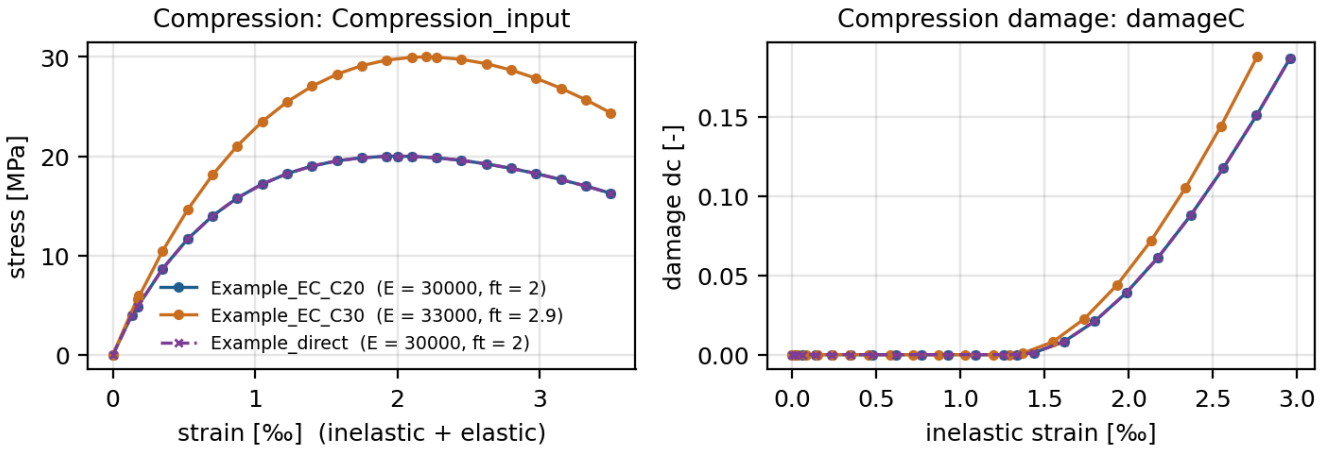


Figure 7. The example materials as the study writes them into the model: compression curve and compression damage. `Example_direct` was made from the same values as `Example_EC_C20` , so the two curves coincide. Run a material script on its own (python) to check new input.

3 Section export (A_1_Export_Auto.py)

3.1 Files and the settings dialog

Copy everything whose name starts with `A_` from `3_export_and_image_export` into the folder with the ODB files (or anywhere, and choose the ODB folder in the dialog):

```
Benchmark_wall\
  A_1_Export_Auto.py              step 3: section export
  A_2_Image_save_view.py          step 4: save a picture (chapter 4)
  A_2_Image_export_Auto.py        step 4: images and curves (chapter 5)
  A_IMPORT_additional_functions\  library - must stay next to the scripts
  SHEAR_key_S_FR05_EX_34272.odb
  SHEAR_key_S_FR05_EX_172699.odb  ...
```

Run `A_1_Export_Auto.py` with *File* > *Run Script*... in Abaqus/CAE or Viewer. The settings dialog opens with the settings of the last export:

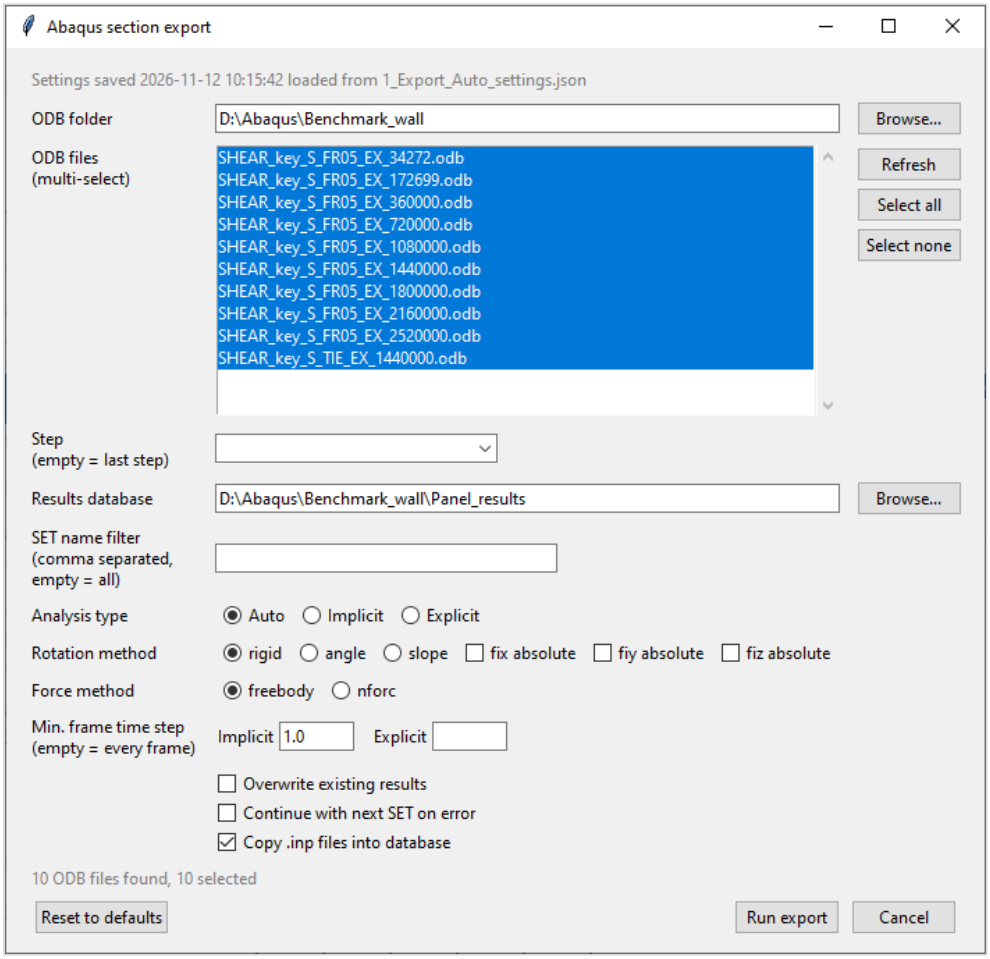


Figure 8. Settings dialog of the section export.

field	meaning
ODB folder, files	all ODB files of the folder are selected; multi-select with Ctrl / Shift
Step	empty = the last step of each ODB (e.g. `HOR`)
Results database	folder of the results (default `Panel_results` next to the script)
SET name filter	only SETs containing one of these texts, e.g. `RU, DPARC`; empty = all
Analysis type	Auto detects implicit / explicit per ODB
Rotation method	**rigid**: rigid-body fit, valid for large rotations; angle, slope: regression plane (moderate / small rotations). *absolute*: orientation instead of the change from the start
Force method	**freebody**: Abaqus free-body cut; **nforc**: sum of NFORC by the script (same result, see 3.4)
Min. frame time step	keep a frame only if it is at least this much step time after the last kept one (implicit default 1.0); empty = every frame
Overwrite, Continue, Copy .inp	re-export existing sections; go on after a failing SET; copy the input file of each job into the database

Run export saves the settings and starts. Sections already in the database are skipped unless *Overwrite* is ticked. Without the dialog (`USE_GUI = False`) the export also runs in batch mode:
`abaqus cae noGUI=A_1_Export_Auto.py` (the 16 sections of the example take 13 s).

3.2 What is computed

For every ODB and every SET, frame by frame:

- **Displacements and rotations** (`Ux Uy Uz fix fiy fiz`) of the section from the displacements `U` of its nodes: the rigid-body motion that fits the nodes best (also correct for large rotations), resolved in the section axes. Rotations are relative (change from the undeformed state) unless *absolute* is ticked.
- **Forces** (`Fx Fy Fz Mx My Mz`) acting on the element set at the cut nodes, in the section axes, about the area-weighted centre of the cut: by an Abaqus free-body cut (default), or by summing the element nodal forces NFORC directly from the ODB.
- Displacements and forces are matched **frame by frame** by the position of the frame in the step (in Abaqus/Explicit the frame id is the increment number, which is not used).

3.3 The results database

The results are text files in one folder per SET and job. The database of the example:

```
Panel_results\
  baza sadzaj liste.txt                    list of the exported jobs
  V_PIER&RU&2&0\
    Attributes.json                        centre and size of the cut
    SHEAR_KEY_S_FR05_FORCE_34272\
      Section_dataIIIIV_PIER&RU&2&0IIIISHEAR_KEY_S_FR05_FORCE_34272.txt
  V_PIER&RU&1&0\  ...                    16 SET folders
```

```
# Section_dataIIIIV_PIER&RU&2&0IIIISHEAR_KEY_S_FR05_FORCE_34272.txt  (first rows of 53)
Time,frameId,Ux,Uy,Uz,fix,fiy,fiz,Fx,Fy,Fz,Mx,My,Mz,
0.0,0,0.0,0.0,-0.0416,0.0,0.0,0.0,0.002,5.348,-34635.2,823195.5,-1552.0,-24.6,
1.0,1,0.0167,0.0,-0.0416,0.0,0.0,0.0,755.6,5.554,-34636.3,823403.7,-1070320.0,15537.6,
2.0,2,0.0591,0.0,-0.0416,0.0,0.0,0.0,2344.1,5.430,-34637.8,823233.0,-3316900.0,48613.2,
# Attributes.json
{"DimensionZ": 0.0, "DimensionX": 1200.0005, "DimensionY": 480.0,
 "CoordZ": 2839.9998, "CoordY": 19.9998, "CoordX": 1800.0002}
```

Values are in the units of the model (here N, mm, rad, Nmm; numbers shortened). *Time* is the step time, *frameId* the position of the frame in the step.

3.4 Checking the results

Three checks, from quick to thorough:

1. **Equilibrium:** the normal force of a horizontal cut through the whole wall equals the vertical load (plus self-weight); top and base shear are equal and opposite (signs, 1.3). Near-zero forces: see 1.4 (the export warns about SETs with elements on both sides of the cut).
2. **Both methods:** export once with *freebody* and once with *nforc* into two databases, or run `section_forces.py` of the library on its own with `"METHOD": "compare"` in its `DEMO_SETTINGS`; it prints both and their differences. Figure 9 shows the result for the example.
3. **Abaqus itself:** in Abaqus/Viewer create a free-body cut from the node set and the element set of the SET (*Tools* > *Free Body Cut* > *Create*...), plot its force over time (*Tools* > *XY Data* > *Create*..., source *Free body*) and compare with the column of the section file. Resolve the cut in the section axes (a datum coordinate system) to compare the components.

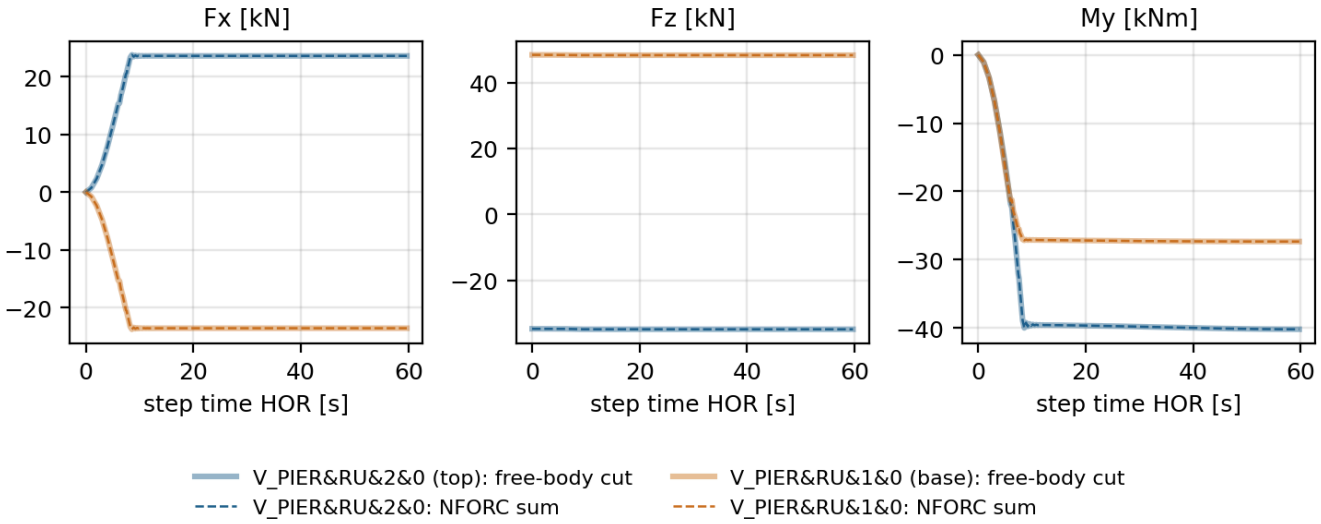


Figure 9. Fx, Fz and My of the top and the base section of the example, from the Abaqus free-body cut (thick) and from the NFORC sum (dashed): all 153 frames agree within 0.001 %.

4 Defining pictures (A_2_Image_save_view.py)

Pictures are set up once in Abaqus/Viewer as usual, and saved as a **view** that the image export (chapter 5) replays for every ODB and frame.

1. Open an ODB in Abaqus/Viewer and set up the picture: camera, section (view) cut, display group (hide or show parts), variable, contour type and limits, deformation scale, visible edges, legend, triad, background.
2. *File* > *Run Script...* > *A_2_Image_save_view.py* (or the Plug-ins button, see below) and fill in the dialog.
3. Repeat for every picture; then run `A_2_Image_export_Auto.py` .

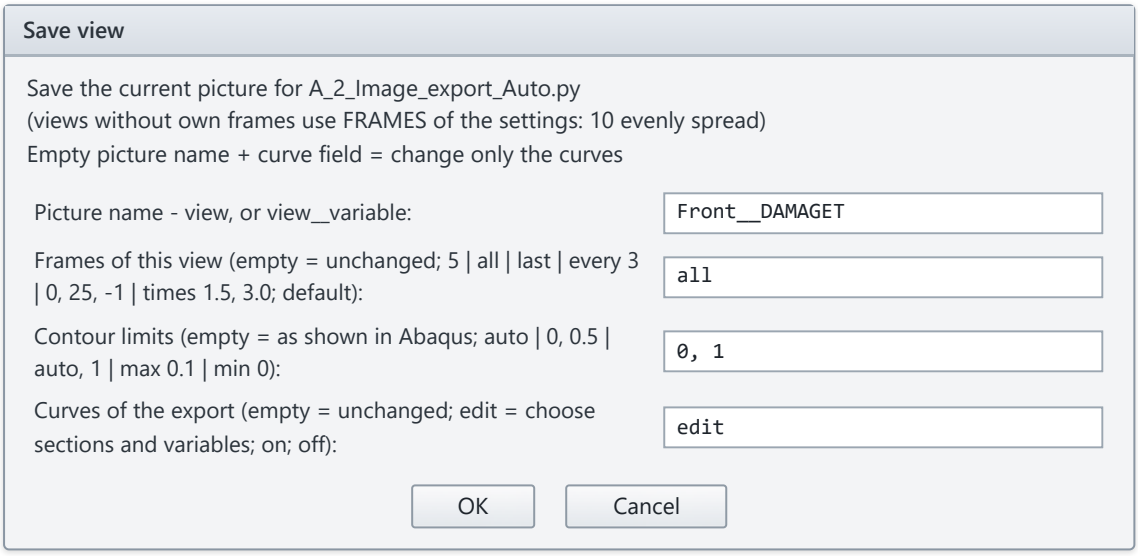


Figure 10. The dialog of `A_2_Image_save_view.py` (Abaqus' own input dialog; texts as in the script).

field	meaning
Picture name	`Front` : view "Front" with the variable shown now; `Front__DAMAGET` : view "Front", variable named "DAMAGET". Saving the same view with another variable adds the variable; camera etc. are updated.
Frames	frames of this view: `5` (evenly spread), `all` , `last` , `every 3` , `0, 25, -1` (positions), `times 1.5, 3.0` (step times), `default` (FRAMES of the settings); empty = unchanged
Contour limits	of this picture (view + variable): empty = as shown in Abaqus (fixed values or automatic are read from the session), `auto` , `0, 0.5` , `auto, 1` , `max 0.1` , `min 0` . Two views of one variable can have different limits.
Curves	`edit` opens the curve dialog (Figure 11), `on` / `off` switches the curves of the export; empty = unchanged

What is stored: camera, section cuts, edges, render style, deformation scale, contour settings, legend, title, triad, background, plot state, variable and limits – read from the session; the display group is taken from this session's `abaqus.rpy` (Abaqus cannot report it otherwise). A preview PNG is saved in `Panel_images/_view_previews` .

The curve dialog

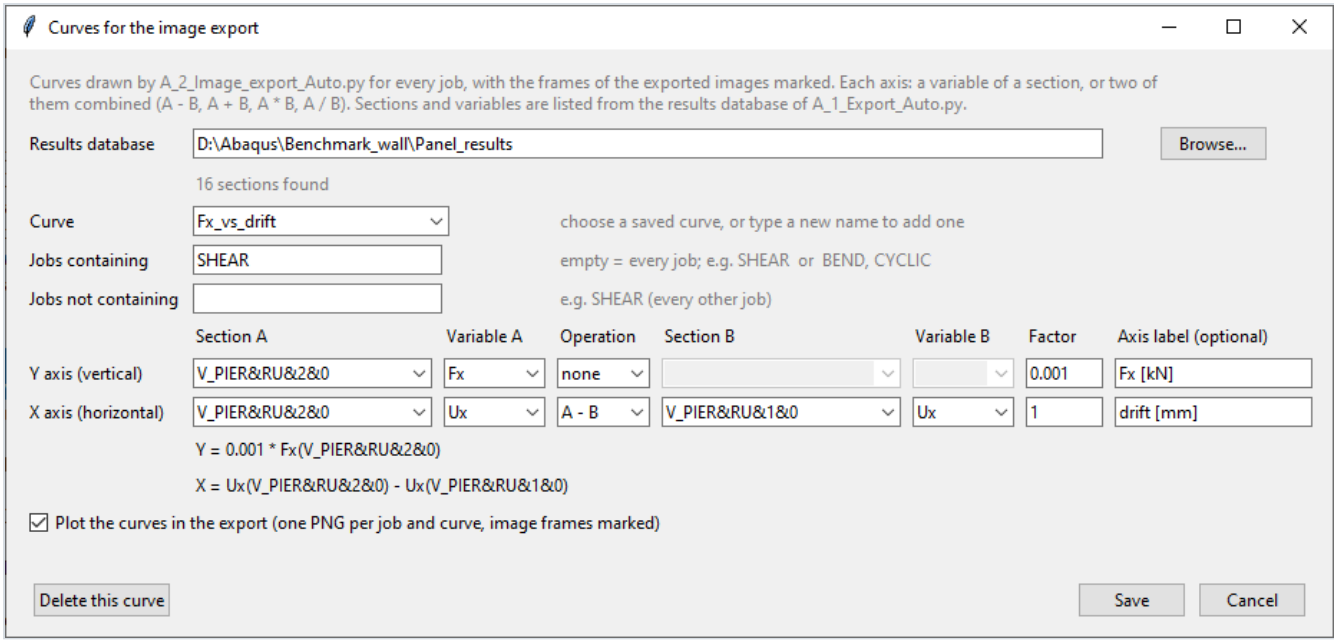


Figure 11. Curve dialog (field *Curves* = `edit`): the sections and variables are listed from the results database of chapter 3. Shown: the curve "Fx_vs_drift" with Y = 0.001 × Fx of the top section (kN) and X = Ux of the top minus Ux of the base section, for every job whose name contains SHEAR.

- Each axis is a variable of one section, or two of them combined: *A* − *B* (e.g. relative displacement of two sections), *A* + *B*, *A* × *B*, *A* / *B* (e.g. Fx/Fz). *Factor* scales the axis (0.001: N → kN, −1: sign); *Axis label* replaces the formula as label.
- *Jobs containing / not containing* choose the jobs a curve is drawn for, e.g. one curve for the SHEAR jobs and another one for all others.
- *Save* stores the curve (and draws a preview for the ODB shown in Abaqus); several curves can be defined.

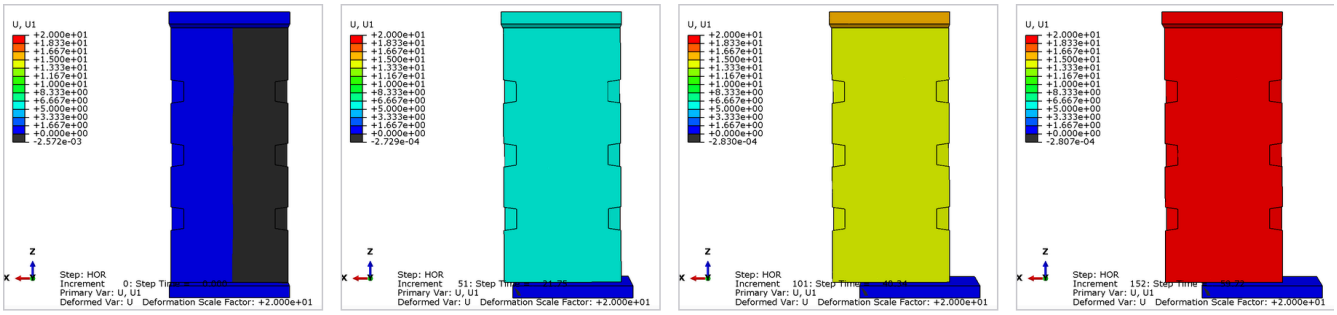
> **Other ways to define pictures.** *Capture*: print every picture with *File* > *Print* to a PNG named `<view>__<variable>`, then run `A_2_Image_export_Auto.py` with `MODE = "capture"`; it reads the views from `abaqus.rpy`. *Plug-ins button*: copy the two files of `optional_abaqus_plugin` to `C:\Users\<name>\abaqus_plugins` and restart Abaqus; *Plug-ins* > *Image export* > *Save current view...* then runs the script of the ODB's folder.

5 Images and curves (A_2_Image_export_Auto.py)

Run `A_2_Image_export_Auto.py` (*File* > *Run Script...*, or `abaqus cae noGUI=A_2_Image_export_Auto.py`). It exports **every ODB × every view × its variables × its frames** and draws the curves of every job. The first lines of its log list what will be done:

```
Views        : Front [default], Front_fixed [last]
Variables    : S_Min_Principal, U1
Frames       : default = 4 evenly spread   |   image size: [1200, 800]
Limits       : Front                      S_Min_Principal    automatic
Limits       : Front                      U1                 min 0.0 (fixed) / max 20.0 (fixed)
Limits       : Front_fixed                S_Min_Principal    min -10.0 (fixed) / max 0.0 (fixed)
Curve        : Fx_vs_drift: Y = 0.001 * Fx(V_PIER&RU&2&0),
                            X = Ux(V_PIER&RU&2&0) - Ux(V_PIER&RU&1&0)  (jobs all)
...
  step HOR: 4 of 153 frames -> [0, 51, 101, 152]
DONE in 0.1 min: 9 new image(s), 0 failed ODB(s)
```

```
Panel_images\
  SHEAR_KEY_S_FR05_FORCE_34272\                one folder per job
    _curve_Fx_vs_drift.png   _curve_Fx_vs_drift.csv
    U1\Front\f0000_t0.png  f0051_t21.7527.png  f0101_t40.3361.png  f0152_t59.7221.png
    S_Min_Principal\Front\...                  <variable>\<view>\f<frame>_t<step time>.png
    S_Min_Principal\Front_fixed\f0152_t59.7221.png
```



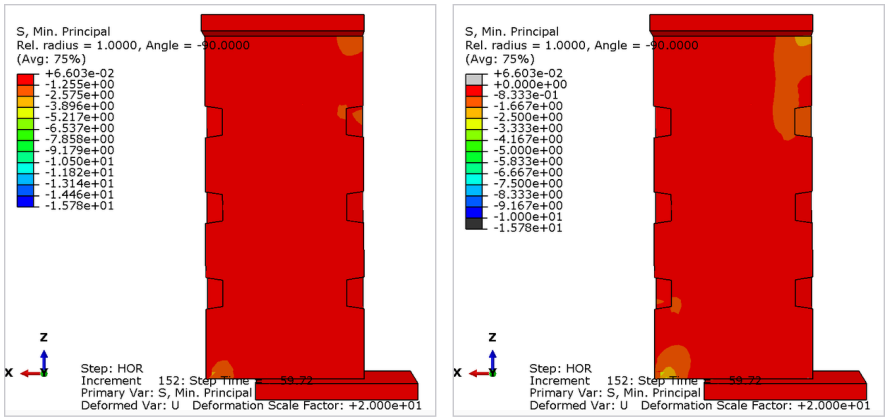
f0000_t0.png
frame 0

f0051_t21.7527.png
frame 51

f0101_t40.3361.png
frame 101

f0152_t59.7221.png
frame 152

Figure 12. The four frames of view "Front", variable U1, with fixed limits 0–20 mm (deformation scale 20): with fixed limits the colours of all frames and jobs can be compared. Values outside the limits are grey (frame 0: −0.003 mm).



view Front
limits automatic

view Front_fixed
limits "-10, 0"

Figure 13. The same frame in two views: automatic limits ("Front") and the limits `-10, 0` ("Front_fixed") – limits belong to the picture (view + variable).

Curves

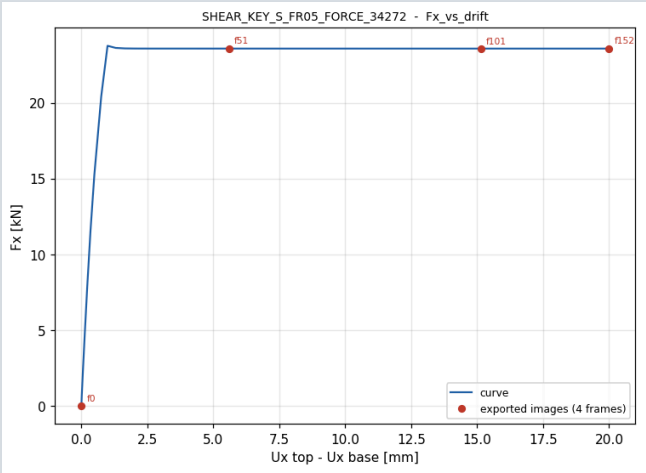


Figure 14. `_curve_Fx_vs_drift.png` of the example: the 4 frames of the images are marked with their number (`f51` = the images `f0051_t21.7527.png`).

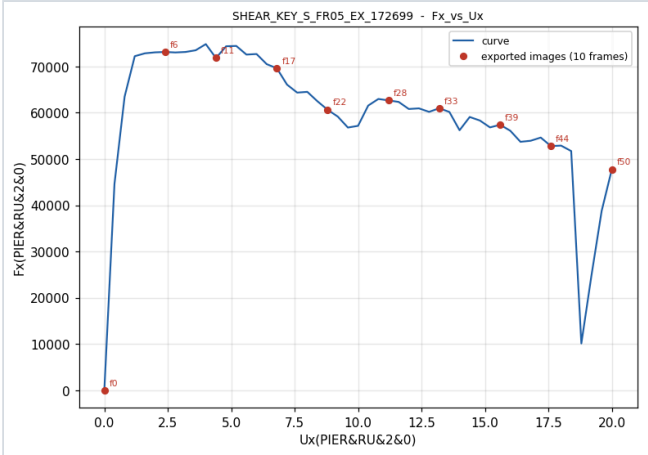


Figure 15. A curve of an explicit analysis of the same wall (`SHEAR_KEY_S_FR05_EX_172699`), 10 image frames.

Next to each PNG a CSV file holds the points and the image frames, for use in a spreadsheet:

```
Time,Ux top - Ux base [mm],Fx [kN],image frames
0.0,0.0,2.00220942497253e-06,f0
1.0,0.01671745441854,0.755562316894531,
2.0,0.0591462291777134,2.34408325195313,
```

Settings

- Views, variables, frames, folders, image size and curves are stored in `2_Image_export_settings.json` ; copy it to another folder to export there with identical settings.
- `FRAMES` (all views without own frames): `{"MODE": "uniform", "COUNT": 10}` , or short text as in the Save view dialog. `IMAGE_SIZE` : e.g. `[1200, 800]` pixels; `None` = as on screen.
- Existing images are kept (`OVERWRITE = False`), so an interrupted export continues; `ZIP_PER_JOB = True` packs every job into one ZIP file (faster to copy).
- `ADD_FRAMES_AROUND_PEAK = True` also exports the peak of the first curve and the points at `BEFORE_PEAK` / `AFTER_PEAK` × peak (e.g. 75 % before and 80 % after the peak).

6 Changing the scripts

The scripts are plain Python 2.7 (Abaqus 2021) and are meant to be adapted. Three typical changes:

6.1 A new parameter study

In `run_parameter_study.py` a study is two functions: the cases (job name ending and value) and the change of the model. Example: a study of the friction coefficient of the contact:

```
def change_friction(model, mu):
    prop = model.interactionProperties[CONTACT_PROPERTY]
    prop.tangentialBehavior.setValues(formulation=PENALTY, table=((mu, ), ),
                                      maximumElasticSlip=FRACTION, fraction=0.005)

def cases_friction():
    return [("_FR%s" % str(mu).replace(".", ""), mu) for mu in (0.4, 0.5, 0.6, 0.7)]

STUDIES = {"contact": (cases_contact, change_contact),
           "material": (cases_material, change_material),
           "friction": (cases_friction, change_friction)}     # then STUDY = "friction"
```

Record the change once in Abaqus/CAE and copy the command from `abaqus.rpy` into the change function.

6.2 A new section type

Section types are one list in `A_IMPORT_additional_functions/section_definitions.py` (the first matching token wins):

```
SECTION_TYPES = [
    SectionType("V_PIER",     ("V_PIER", "V_SPEC"), "volume", "PIER",     True),
    ...
    # new: SETs named V_WALL&...
    SectionType("V_WALL",     ("V_WALL",),          "volume", "PIER",     True),
]
#   name, tokens in the SET name, geometry ("volume" | "shell" | "point"),
#   axes ("PIER" | "SPANDREL" | None), forces exported (True / False)
```

Run the module on its own (`python section_definitions.py`) to see how example names are classified. Also add the token to `SECTION_TOKENS` of `prepare_model_for_export.py` so that the check lists the new SETs.

6.3 Your own post-processing of the results database

The results database is plain text, so any Python (or a spreadsheet) can process it, without Abaqus. Example: the peak shear of the top section and the displacement at the peak, for every job:

```
import os, glob, csv
db, section = r"D:\Abaqus\Benchmark_wall\Panel_results", "V_PIER&RU&2&0"
for path in sorted(glob.glob(os.path.join(db, section, "*", "Section_data*.txt"))):
    rows = list(csv.DictReader(open(path)))
    peak = max(rows, key=lambda r: abs(float(r["Fx"])))
    job = os.path.basename(os.path.dirname(path))
    print("%-32s Fx max = %8.1f kN at Ux = %6.2f mm (t = %s)" % (
        job, float(peak["Fx"]) / 1000, float(peak["Ux"]), peak["Time"]))
```

```
SHEAR_KEY_S_FR05_FORCE_34272     Fx max =     23.8 kN at Ux =   0.99 mm (t = 8.5625)
```

Inside the image export, curves (chapter 4) cover most needs; `pushover_curve.py` reads the database in the same way and can be extended (e.g. a new operation in `OPERATIONS`).

6.4 Where to find what

module (A_IMPORT_additional_functions)	contents
`odb_tools.py`	opening ODBs, frame selection (frame sync), sets, field values, logging
`section_definitions.py`	SET name → section type and axes
`section_geometry.py`	regression plane, rigid-body fit, rotations, cut centroid
`section_displacements.py`	displacements and rotations of the sections
`section_forces.py`	free-body cut and NFORC forces (stand-alone: compare both)
`database_io.py`	results database (folders, file names, columns)
`export_settings.py`, `export_gui.py`	remembered settings, settings dialog
`image_export.py`, `rpy_views.py`, `view_snapshot.py`	replaying views, reading `abaqus.rpy`, snapshot of the session
`pushover_curve.py`	curves from the results database, curve dialog

> **After changing a library file** restart Abaqus/CAE or Viewer: Abaqus keeps imported modules for the whole session. (The scripts only reload library modules that were imported from another folder.)

A Troubleshooting

message or symptom	cause and remedy
Folder A_IMPORT_additional_functions not found next to this script	copy the folder together with the A_ scripts (3.1)
No .odb files found in ...	choose the ODB folder in the dialog, or set `ODB_FOLDER`
WARNING ... elements on BOTH sides of the cut, or a section with forces near zero	the element set holds elements of both sides of the cut: make it again with the elements of one side (1.4, 1.5)
Forces of two sections have opposite signs	their element sets lie on different sides of the cut (1.3)
SET ... must look like TYPE&LABEL&INDEX&ANGLE / INDEX and ANGLE must be numbers	rename the SET (1.2); helper sets must not contain PIER, SPEC, SPANDREL, DRIFT or SRIFT
Abaqus seems to hang during an export	open the `*_log.txt` file next to the script: it shows the current ODB, SET and frame
Model(s) not in the open .cae / no model 'X'	model name or wildcard does not match (2.1); names are listed in the message
model ... has no load VERT_FORCE	set `LOAD_NAME` / `LOAD_STEPS` of `run_pushover_jobs.py` to the names of your model
the dialog could not be shown (noGUI)	batch runs (`abaqus cae noGUI=...`) need `ASK = False` / `USE_GUI = False`
Curve: section ... is not in the database	run the section export first (chapter 3), or choose the results database in the curve dialog
An image looks different from the screen	display group or a setting was not stored: save the view again after setting it up; check the preview in `Panel_images/_view_previews`
A change in a library file has no effect	restart Abaqus (6.4)

B Video tutorials and further reading

The videos were recorded with the **earlier version** of the scripts: the modelling (SETs, naming, local axes) is the same; script names and input differ (no `Input_file.txt` any more – the scripts have an INPUT block, dialogs and remembered settings).

video	link
How the scripts work – overview, from a new model to extracted results	https://youtu.be/g9gw_EXEesM
Creation of the benchmark model	https://youtu.be/stwDfm0-Hc0
Cross-section naming and local axes (chapter 1)	https://youtu.be/6BFI6Eegp64
Export script (chapter 3)	https://youtu.be/nRnF_rlyfsg
Displacement and force conversion modification demo	https://youtu.be/nNw8NbOuac0
Automated job running (chapter 2)	https://youtu.be/iR5NVptAlkw
Adding a custom post-processing function (chapter 6)	https://youtu.be/oXEkcqDdTa8
Entire playlist	https://www.youtube.com/playlist?list=PLjdnR7Fv3RGjCVFS0ykvkd6-2jo0IHgPl

In the repository

- `README.md` of the repository and of every step folder: short reference of the scripts.
- `docs/flowchart` : flowchart and short report of steps 3 and 4 (PDF, PNG, SVG, HTML, Visio, draw.io).
- `readme.txt` : the texts of the earlier README files.
- The INPUT block and the description at the top of every script.

This tutorial: `docs/tutorial/source/tutorial.html` ; the PDF is rebuilt with `python build_pdf.py` in that folder (Microsoft Edge and PyMuPDF).